

Problem

Design a circuit that provides a relationship between output voltage v_o and input voltage v_s such that $v_o = 12v_s - 10$. Two op amps, a 6-V battery, and several resistors are available.

Step-by-step solution

Step 1 of 3

Consider that two operational amplifiers, a 6 V battery, and several resistors are available. Design a circuit that provides the following relation.

$$v_o = 12v_s - 10$$

Here,

The output voltage is v_o , and

The input voltage is v_s .

Step 2 of 3

Estimate the circuit using the available data as follows:

Input voltage, v_s is being multiplied by the factor 12.

An inverting amplifier amplifies a signal and reverses the polarity.

Use an inverting amplifier in the circuit design to produce an output, v_1 as follows:

$$v_1 = -12v_s$$

The terms $12v_s$ and 10 are being added in the equation.

A summing amplifier combines several inputs and produces an output as a weighted sum of the inputs.

Use a summing amplifier in the circuit design to produce an output, v_o as follows:

$$\begin{aligned} v_o &= - \left[(-12v_s) + \left(\frac{5}{3} \right) (6 \text{ V}) \right] \\ &= 12v_s - 10 \end{aligned}$$

Step 3 of 3

The following is the circuit design:

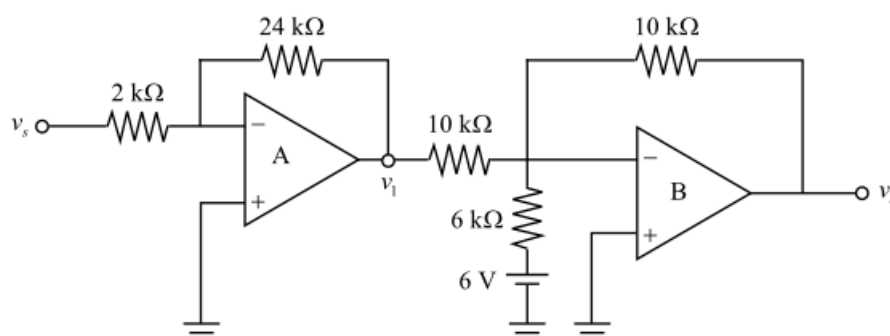


Figure 1

